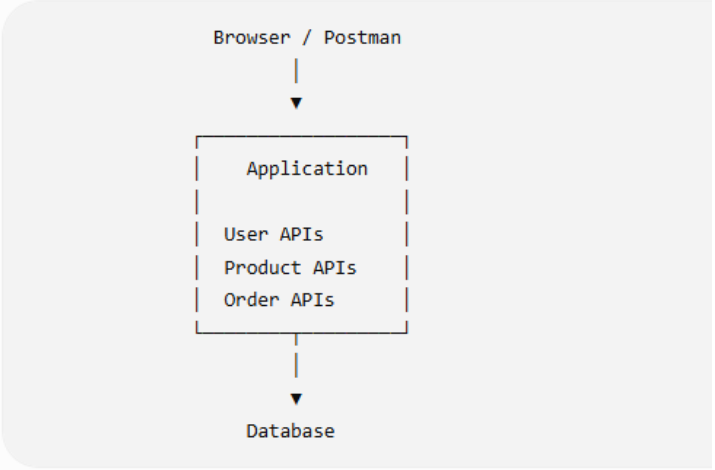
Planning Our Microservices Transofrmation

1. Current Architecture - Monolithic

The application currently looks like this:



The important point is that:

- User APIs
- Product APIs
- Order APIs

are all part of the **same application**.

And the entire application communicates with the database.

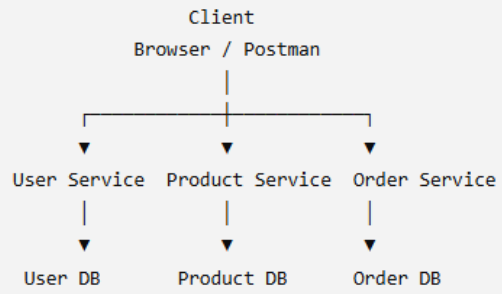
2. What are we changing?

We are going to split the existing application into **three microservices**.

Current:

```
User + Product + Order
        ↓
   One Application
        ↓
    One Database
```

New:

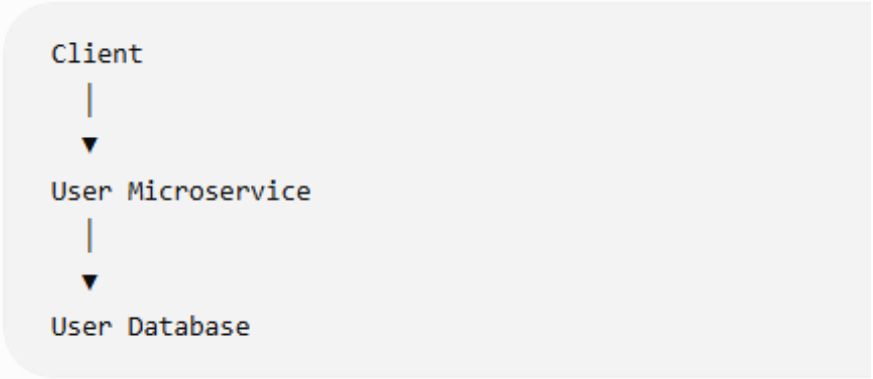


So:

One monolithic application → Multiple microservices

3. User Microservice

The User functionality becomes its own microservice.



The User database contains data related to users.

For example:

- User information
- User-related data
- Other data belonging to the User domain

4. Product Microservice

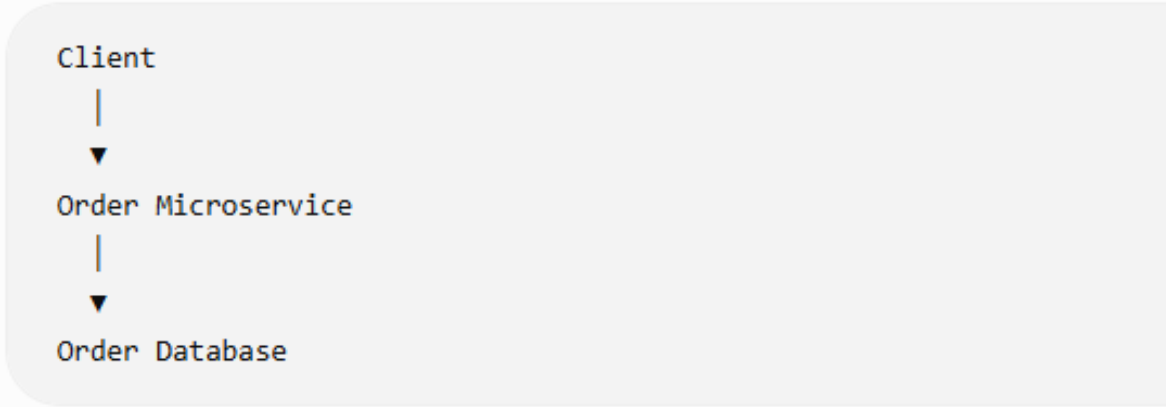
Similarly, Product functionality becomes a separate microservice.



Product-related information resides in the Product database.

5. Order Microservice

Order functionality becomes another independent microservice.



Order-related information resides in the Order database.

6. Database Per Microservice

One of the major architectural changes in this lecture is the move from:

```
            One Database
            /     |     \
          User   Product  Order
```

to:

```
User Service    → User Database

Product Service → Product Database

Order Service   → Order Database
```

This is commonly described as:

> **Database per service**

The idea is that each microservice owns the data associated with its business domain.

7. Each Service can use a Different Database Technology

```
User Team
    ↓
User Service
    ↓
Database hosted on Google Cloud


Product Team
    ↓
Product Service
    ↓
Database hosted on AWS


Order Team
    ↓
Order Service
    ↓
Another database/hosting option
```

8. Team Autonomy

This architecture also allows different teams to work independently.

For example:

```
User Team
    ↓
User Microservice
    ↓
User Database


Product Team
    ↓
Product Microservice
    ↓
Product Database


Order Team
    ↓
Order Microservice
    ↓
Order Database
```

Each team can focus primarily on its own business functionality.

9. Monolithic Vs Microservice

Monolithic	Microservices
One application	Multiple services
User/Product/Order in same application	Separate User/Product/Order services
One database	Separate database for each service
Tightly coupled application	More independent services
Less team autonomy	Greater team autonomy
Technology decisions largely centralized	Services can potentially choose different technologies